Athlete Load & Injury Risk Tracking System

by

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1. ABSTRACT

The Athlete Load and Injury Risk Tracking System is a database-driven application designed to offer a centralized and efficient solution for managing athlete health and performance information. The system aims to simplify the process of recording training workloads, injury histories, recovery metrics, and physiotherapy sessions, particularly within amateur sports settings such as colleges and training academies. By transitioning from fragmented and manual tracking methods to a structured relational database, sports organizations can achieve greater accuracy, consistency, and ease of analysis in athlete management.

The system comprises several key functional modules, including athlete and coach management, training load monitoring, injury logging, recovery tracking, and physiotherapy session management. It also supports secure and logical associations such as linking athletes to multiple coaches or mapping specific treatment sessions to ongoing injuries thereby reflecting real-world relationships within sports environments. All underlying data is stored and maintained in a relational database normalized up to the Boyce-Codd Normal Form (BCNF) to ensure integrity, eliminate redundancy, and promote efficient data handling.

Overall, this project serves as a practical demonstration of advanced database management principles, including entity-relationship modeling, relational schema design, and SQL Data Definition Language (DDL) implementation. It provides a robust and scalable system that enhances collaboration among athletes, coaches, and medical staff while ensuring precise and reliable management of health and performance data.

1. INTRODUCTION

In amateur sports environments such as colleges and training academies, monitoring athlete performance and health is essential for supporting long-term athletic development. However, as the number of athletes, coaches, and medical staff increases, the management of related data becomes significantly more complex. Maintaining clear relationships between daily training intensity, recovery indicators such as sleep quality and fatigue, and possible injury risks requires a systematic and well-organized method of record-keeping.

In many existing sports programs, data is still collected and maintained manually, often resulting in records that are fragmented, inconsistent, and difficult to analyze over time. At the same time, coaches and physiotherapists must handle detailed and interconnected information, such as associating a specific injury, like a hamstring strain, with the corresponding treatment sessions prescribed for recovery. In the absence of a structured system, this scattered approach can negatively affect both injury management and training optimization.

To address these challenges, the Athlete Load and Injury Risk Tracking System has been developed as an integrated relational database solution. The system is designed to organize and manage critical processes, including training workload assessment, daily wellness monitoring, and the identification of patterns that may indicate athlete fatigue or burnout. By replacing traditional paper-based records with a formalized SQL-based database, the system aims to improve administrative efficiency while also supporting better decision-making and healthier outcomes for athletes.

3. EXISTING WORK

In existing amateur sports environments:

- Athlete performance and health data are typically recorded using basic spreadsheets or paper-based logs.
- Athlete records are frequently fragmented and inconsistent across different departments.
- There is no structured, easily accessible history of an athlete's past injuries and recovery timelines.
- Staff face significant difficulty in tracking and analyzing an athlete's training workload over extended periods.
- There is poor linkage and communication between the records of sustained injuries and the subsequent physiotherapy sessions meant to treat them.

These limitations result in a high risk of data loss and duplication, ultimately making it difficult to prevent injuries and optimize athlete performance effectively.

4. PROPOSED SYSTEM

The proposed system provides:

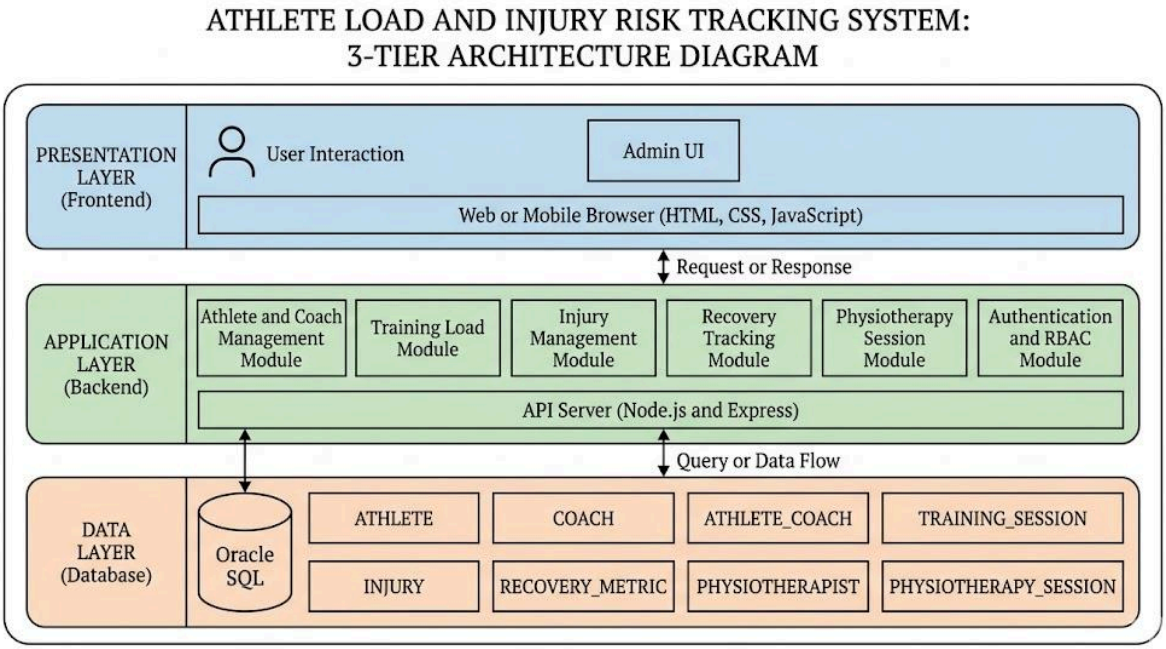
- Centralized athlete records.
- Automated workload and injury-risk monitoring using recent training and recovery data.
- Dashboard-based insights with key metrics and a unified athlete timeline.

Advantages:

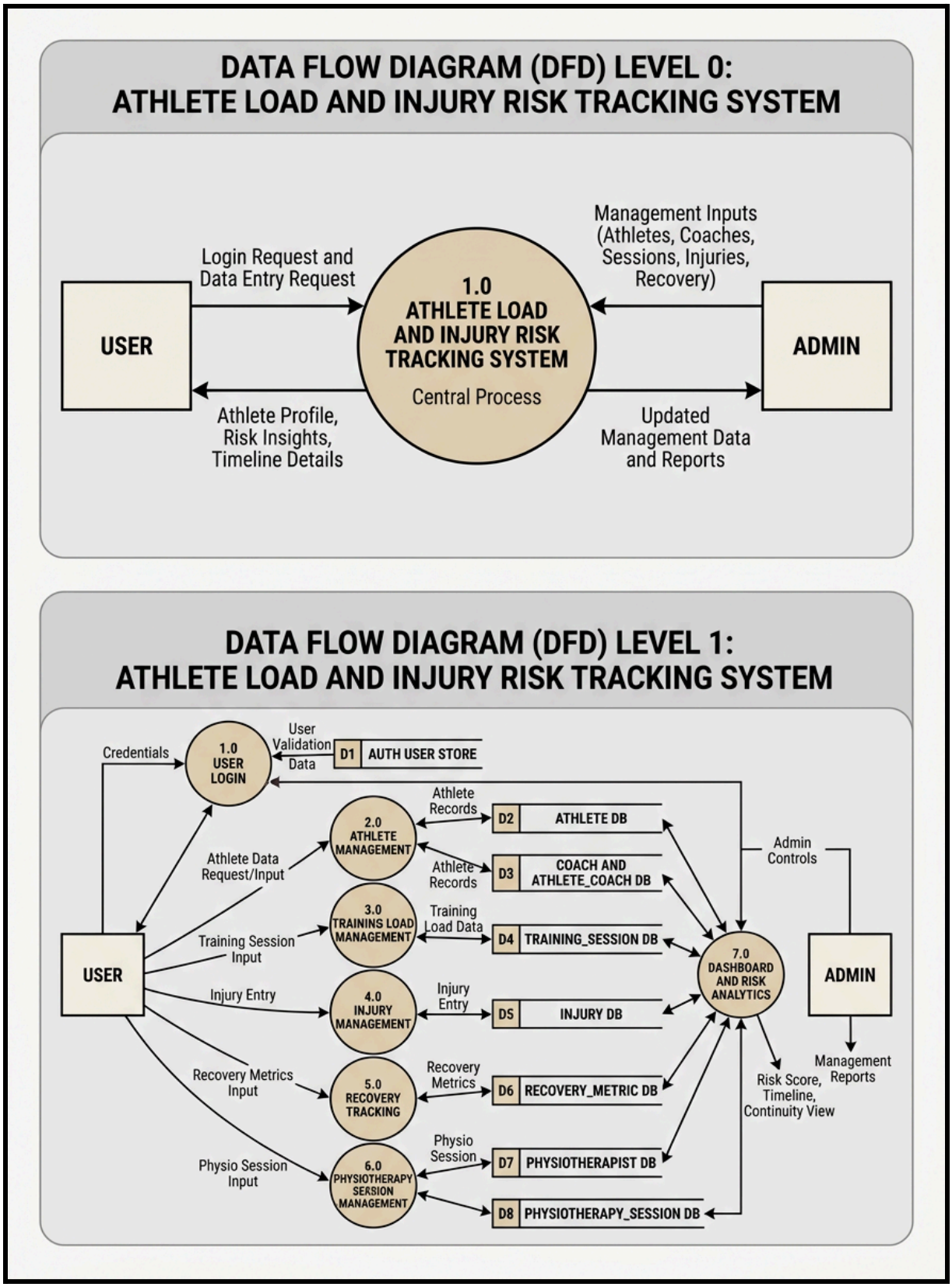
- User-friendly interface for quick data entry, review, and monitoring.
- Reduced manual effort through digitized record management.
- Improved data accuracy, consistency, and communication across departments.
- Faster decision-making with real-time analytics and secure, scalable database storage.

5. SYSTEM ARCHITECTURE / DESIGN

3-Tier Architecture



Data Flow Diagram



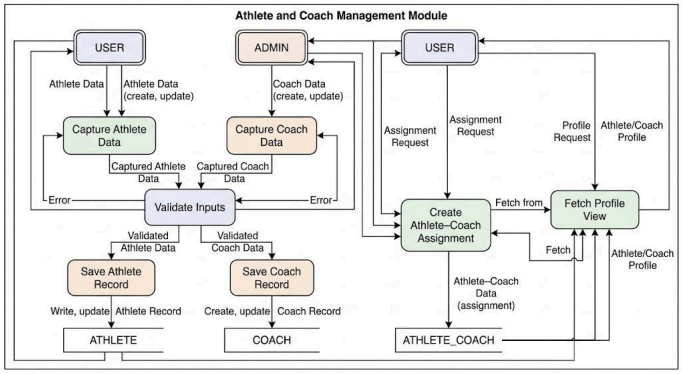
6. TECHNOLOGY STACK

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: Oracle SQL DB
- Tools: GitHub, VS Code

7. WORKING MODULES

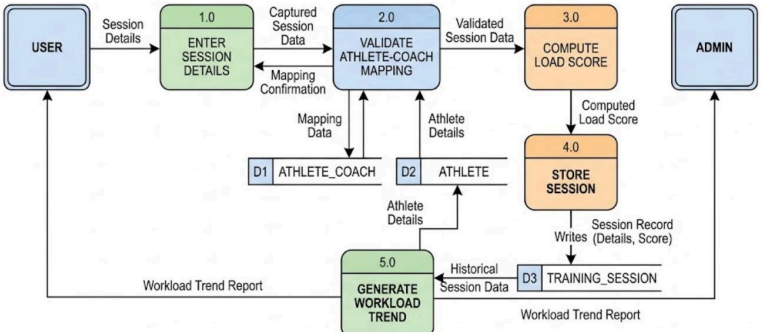
1. Athlete and Coach Management Module

- Manages athlete and coach records
- maps coaches to athletes for structured supervision history
-



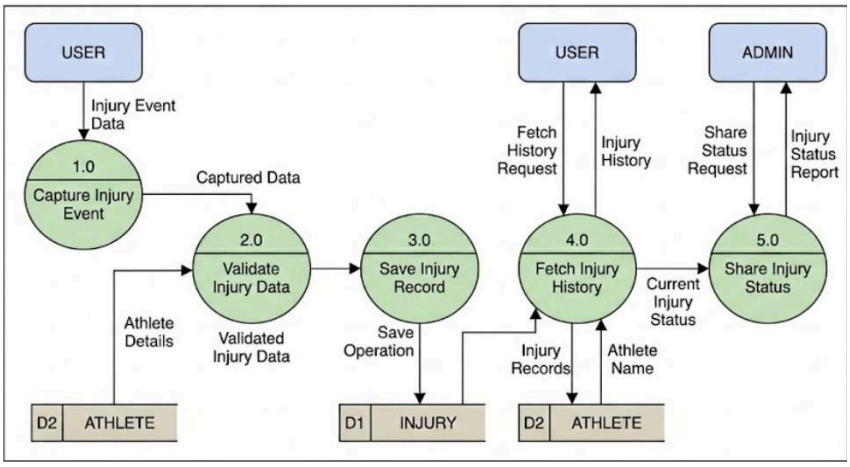
2. Training Load Module

- Records training sessions and computes workload trends for performance and risk monitoring.



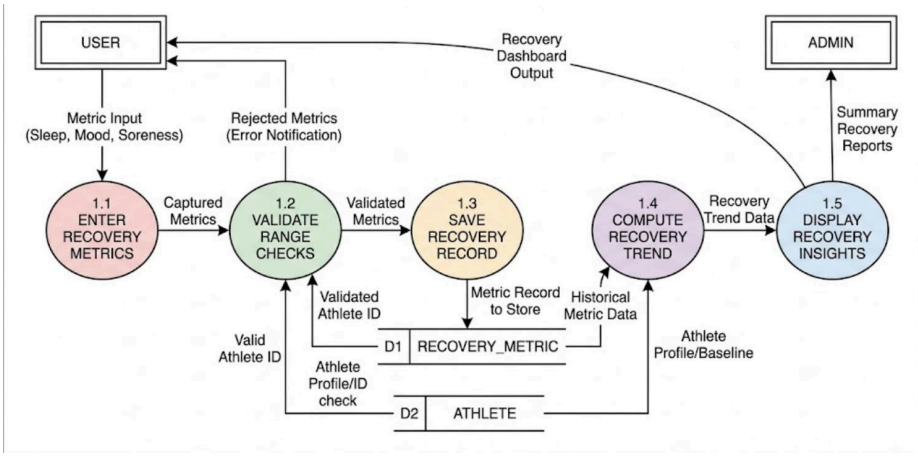
3. Injury Management Module

- Captures injury incidents and maintains athlete-wise injury history for continuity of care.



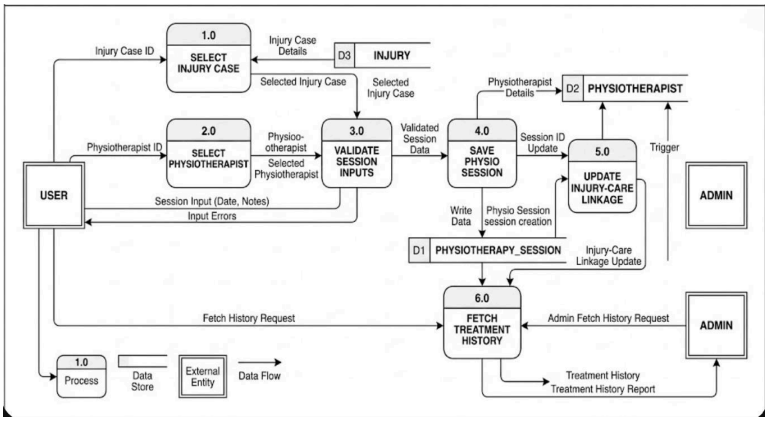
4. Recovery Tracking Module

- Tracks fatigue, soreness, and sleep metrics to monitor recovery progression over time.



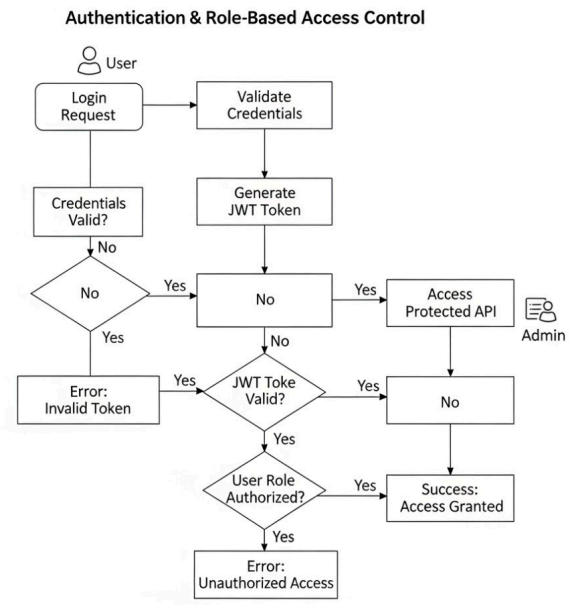
5. Physiotherapy Session Module

- Links injuries with physiotherapy sessions to ensure treatment traceability and rehabilitation tracking



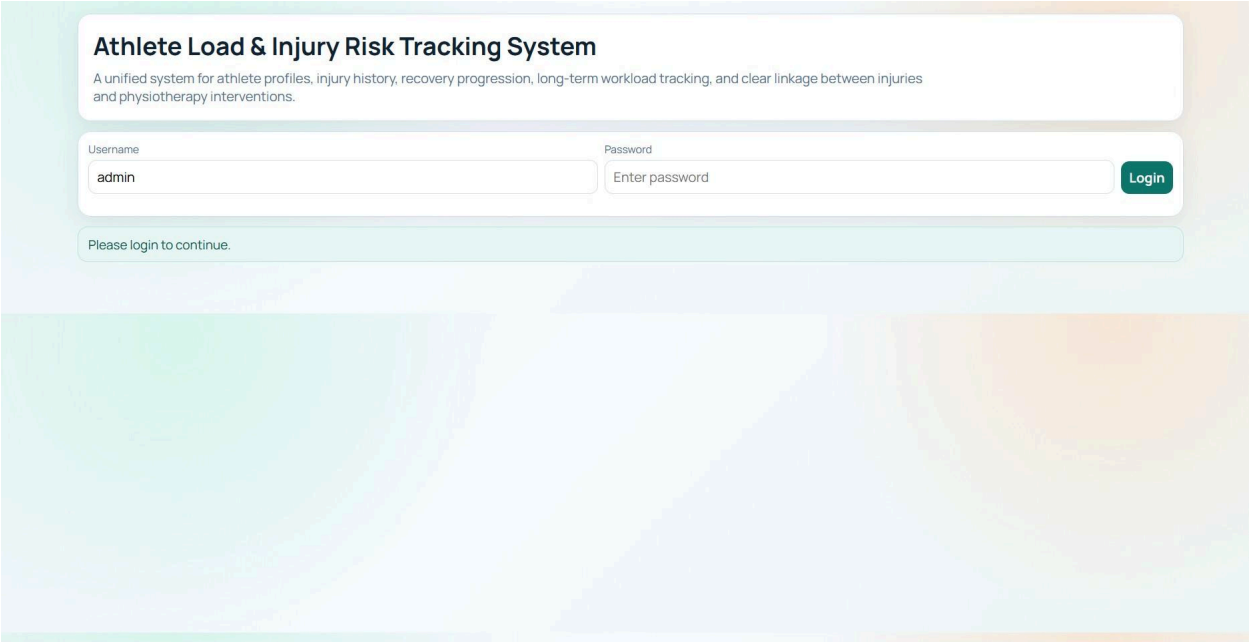
6. Authentication and RBAC Module

- Authenticates users and enforces role-based access control for secure API operations.

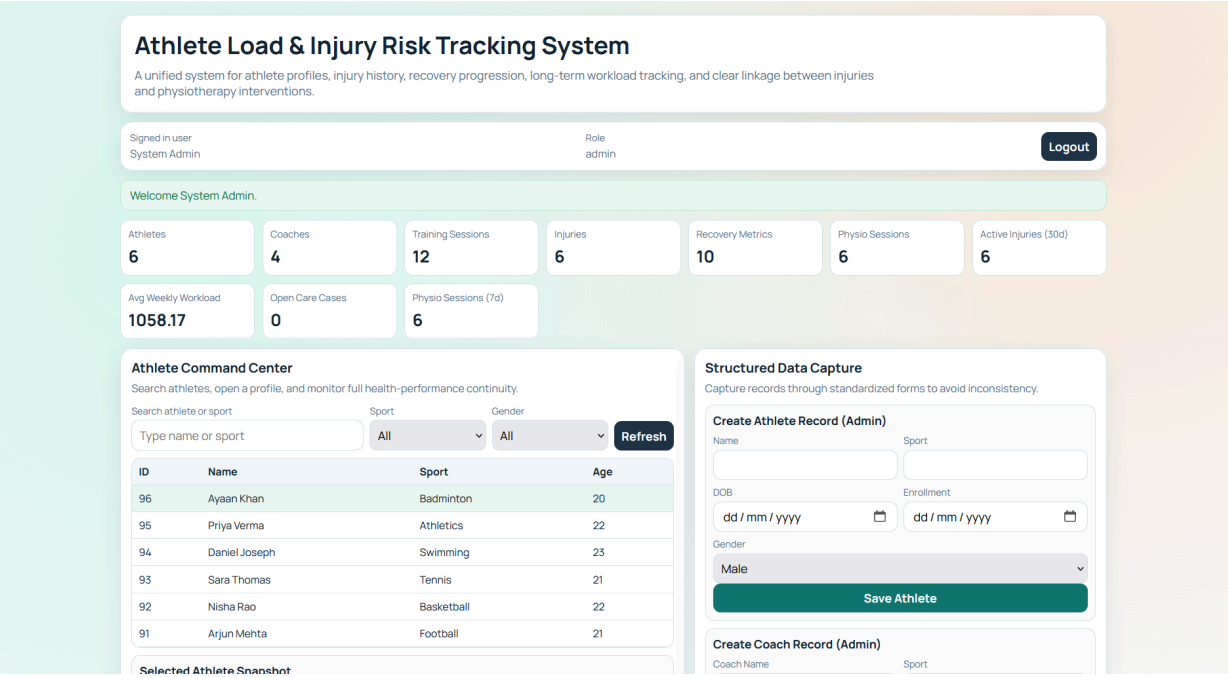


8. OUTPUT SCREENSHOTS

Login Page



Admin Panel


Athlete Load & Injury Risk Tracking System
A unified system for athlete profiles, injury history, recovery progression, long-term workload tracking, and clear linkage between injuries and physiotherapy interventions.
Signed in user
System Admin
Role
admin
Logout
Welcome System Admin.
Athletes
6
Coaches
4
Training Sessions
12
Injuries
6
Recovery Metrics
10
Physio Sessions
6
Active Injuries (30d)
6
Avg Weekly Workload
1058.17
Open Care Cases
0
Physio Sessions (7d)
6
Athlete Command Center
Search athletes, open a profile, and monitor full health-performance continuity.
Search athlete or sport
Type name or sport
Sport
All
Gender
All
Refresh
ID Name Sport Age
96 Ayaan Khan Badminton 20
95 Priya Verma Athletics 22
94 Daniel Joseph Swimming 23
93 Sara Thomas Tennis 21
92 Nisha Rao Basketball 22
91 Arjun Mehta Football 21
Selected Athlete Snapshot
Structured Data Capture
Capture records through standardized forms to avoid inconsistency.
Create Athlete Record (Admin)
Name
Sport
DOB
dd / mm / yyyy
Enrollment
dd / mm / yyyy
Gender
Male
Save Athlete
Create Coach Record (Admin)
Coach Name
Sport

Athlete Command Center

Athlete Command Center

Search athletes, open a profile, and monitor full health-performance continuity.

Search athlete or sport: Type name or sport | Sport: All | Gender: All | Refresh

ID	Name	Sport	Age
96	Ayaan Khan	Badminton	20
95	Priya Verma	Athletics	22
94	Daniel Joseph	Swimming	23
93	Sara Thomas	Tennis	21
92	Nisha Rao	Basketball	22
91	Arjun Mehta	Football	21

Selected Athlete Snapshot

#93 - Sara Thomas
Sport: Tennis | Gender: Female | Age: 21
Enrollment: 2025-01-15

Coach Mapping
- Vikram Das (Racquet Sports) since 2025-01-15

Workload Trend (Last 60 Days)

2026-04-02	546
2026-03-30	490

Injury to Physiotherapy Linkage

Injury	Date	Severity	Physio Sessions
Ankle Sprain	2026-03-31	Low	2026-04-01: Mobility and Taping (Dr. Rohan Pillai)

Recovery and Risk Monitor

Low Risk (Score 28)

Workload: 1036, Fatigue: 5, Soreness: 4, Sleep: 7, Active injuries (30d): 1

Date	Fatigue	Soreness	Sleep
2026-04-02	4	3	8
2026-03-31	6	5	6

Unified Timeline (logs)

Cross-Department Unified Timeline

All critical events in one chronological feed for better communication and decisions.

Date	Type	Athlete	Details
2026-04-04	TRAINING	Nisha Rao	Load 432 (72 min, intensity 6)
2026-04-04	PHYSIO	Ayaan Khan	Preventive Mobility Session by Dr. Kavya Singh
2026-04-04	PHYSIO	Nisha Rao	Preventive Mobility Session by Dr. Kavya Singh
2026-04-04	RECOVERY	Ayaan Khan	Fatigue 5, Soreness 4, Sleep 7
2026-04-04	RECOVERY	Nisha Rao	Fatigue 5, Soreness 4, Sleep 7
2026-04-04	INJURY	Ayaan Khan	General Overuse Alert [Low]
2026-04-04	INJURY	Nisha Rao	General Overuse Alert [Low]
2026-04-04	TRAINING	Ayaan Khan	Load 432 (72 min, intensity 6)
2026-04-03	PHYSIO	Priya Verma	Strength Reconditioning by Dr. Kavya Singh
2026-04-02	TRAINING	Sara Thomas	Load 546 (78 min, intensity 7)
2026-04-02	PHYSIO	Daniel Joseph	Therapeutic Ultrasound by Dr. Leena Das
2026-04-02	RECOVERY	Sara Thomas	Fatigue 4, Soreness 3, Sleep 8
2026-04-02	RECOVERY	Ayaan Khan	Fatigue 5, Soreness 4, Sleep 8
2026-04-02	INJURY	Priya Verma	Knee Tendonitis [High]
2026-04-02	TRAINING	Priya Verma	Load 704 (88 min, intensity 8)
2026-04-01	TRAINING	Ayaan Khan	Load 680 (85 min, intensity 8)
2026-04-01	PHYSIO	Sara Thomas	Mobility and Taping by Dr. Rohan Pillai
2026-04-01	RECOVERY	Priya Verma	Fatigue 8, Soreness 7, Sleep 5
2026-04-01	RECOVERY	Daniel Joseph	Fatigue 6, Soreness 5, Sleep 7
2026-04-01	INJURY	Daniel Joseph	Shoulder Fatigue [Moderate]
2026-04-01	TRAINING	Nisha Rao	Load 200 (50 min, intensity 4)
2026-03-31	TRAINING	Priya Verma	Load 300 (60 min, intensity 5)
2026-03-31	PHYSIO	Arjun Mehta	Manual Therapy by Dr. Kavya Singh
2026-03-31	RECOVERY	Sara Thomas	Fatigue 6, Soreness 5, Sleep 6

Data Entry

Create Athlete Record (Admin)

Name | Sport

DOB: dd / mm / yyyy | Enrollment: dd / mm / yyyy

Gender: Male

Save Athlete

Create Coach Record (Admin)

Coach Name | Sport

Phone | Email

Save Coach

Assign Coach to Athlete

Athlete: Select athlete | Coach: Select coach

Start Date: dd / mm / yyyy

Assign

Record Training Workload

Athlete: Select athlete | Coach: Select coach

Session Date: dd / mm / yyyy | Duration (min)

Intensity (1-10)

Save Training Session

Record Injury Event

Athlete: Select athlete | Severity: Low

Injury Type | Injury Date: dd / mm / yyyy

Description

Save Injury

Record Recovery Metric

Athlete: Select athlete | Metric Date: dd / mm / yyyy

Fatigue | Soreness

Sleep Quality

Save Recovery Metric

Create Physiotherapist Record (Admin)

Name | Specialization

Phone | Email

Save Physiotherapist

Record Physiotherapy Session

Injury: Select injury | Physiotherapist: Select physiotherapist

Session Date: dd / mm / yyyy | Duration (min): 1

Treatment Type

Save Physio Session

9. CONCLUSION

The Athlete Load and Injury Risk Tracking System provides an efficient and user-friendly solution for managing athlete health and performance records in a centralized way. It reduces fragmented data entry by combining athlete profiles, training sessions, injury records, recovery metrics, and physiotherapy session linkage into one integrated platform.

The system is supported by a structured relational database that maintains consistency through primary and foreign key relationships. It also includes role-based access control and administrative capabilities for managing core entities such as athletes, coaches, assignments, and physiotherapists.

Overall, the project demonstrates the practical application of database management concepts to a real-world sports operations problem. It improves record continuity, supports data-driven decisions, and enhances operational efficiency through automation and structured design.

10. REFERENCES

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11. COMPLETE IMPLEMENTATION

1. **Project Structure**
    - **Frontend**: Handles user interaction for login, athlete search, profile viewing, workload tracking, injury and recovery visualization, and data capture forms.
    - **Backend**: Processes API requests, applies validation, enforces role-based access control, computes insights and risk metrics, and manages business logic.
    - **Database**: Stores and relates all core entities such as athletes, coaches, training sessions, injuries, recovery metrics, physiotherapists, physiotherapy sessions, and athlete-coach assignments.
    -
2. **Database Implementation**

    The database is implemented using a relational Oracle schema. Entities include Athlete, Coach, Athlete_Coach, Training_Session, Injury, Recovery_Metric, Physiotherapist, and Physiotherapy_Session. Primary and foreign key constraints are defined to maintain relationships and integrity. SQL operations are used for insertion, retrieval, filtering, and reporting.

3. **Core Functionalities**
    - **Authentication and Access Control:** Supports secure login and role-based permissions for authorized operations.
    - A**thlete and Coach Management:** Maintains athlete records, coach records, and assignment history.
    - **Training Load Management:** Captures session duration and intensity, and stores workload records.
    - **Injury Management**: Records injury type, severity, date, and athlete linkage.
    - **Recovery Tracking**: Captures fatigue, soreness, and sleep quality to monitor recovery patterns.
    - **Physiotherapy Session Management**: Links treatment sessions with injury cases and physiotherapists.

- **Dashboard and Timeline**: Provides summary KPIs, risk monitoring, care linkage visibility, and chronological event tracking.

4. **Admin Operations**
    - The admin functionality supports operational control of the system, including:
    - Creating and updating athlete, coach, and physiotherapist records.
    - Managing athlete-coach mappings and monitoring data quality.
    - Reviewing system-wide metrics, injury cases, and continuity data across modules.

5. **System Workflow**
    - User logs into the system with authorized credentials.
    - System loads dashboard summary, lookups, and athlete records.
    - User selects an athlete to view profile continuity and risk indicators.
    - Training, injury, recovery, and physiotherapy records are captured through structured forms.
    - Backend validates and stores data in Oracle, then updates insights and timeline views.
    - System presents updated risk and continuity outputs for operational decision-making.
6. **Integration**

    All modules are integrated to ensure seamless communication between frontend, backend, and database. The system ensures real-time data handling and efficient processing of user requests.